

unite: Unified liNe Integration Turbo Engine

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Summary

Astronomical spectroscopy, whereby light from celestial sources is dispersed into its constituent wavelengths (colors), is a cornerstone of modern astrophysical research. In particular, spectral lines arising from atomic and molecular transitions in astrophysical gas encode fundamental physical properties such as redshift, chemical composition, temperature, density, and kinematics, making the flexibility, speed, and accuracy of spectral line-fitting tools directly impactful on the scientific return of spectroscopic observations.

unite (Unified liNe Integration Turbo Engine) is a Python package for fast and accurate Bayesian inference of spectral line and continuum models with flexible configurations from one or more spectra simultaneously. It is built primarily on JAX ([Bradbury et al., 2018](#)), NumPyro ([Bingham et al., 2019](#); [Phan et al., 2019](#)), and Astropy ([Astropy Collaboration et al., 2013, 2018, 2022](#)).

Statement of need

Making the most of the growing volume and diversity of spectroscopic observations necessitates joint analysis across all available data to fully exploit their collective constraining power and deliver accurate measurements with representative uncertainties. This requires enforcing shared astrophysical parameters while accounting for and propagating systematic uncertainties in instrument-specific calibrations. An additional challenge arises when the data are undersampled, as evaluating the model at pixel centers rather than integrating over the pixel domain introduces systematic errors in recovered line shapes, yet existing tools that account for this typically do so via computationally expensive supersampling and convolution, limiting their scalability to large datasets and heterogeneous, multi-spectrograph configurations.

The Near-Infrared Spectrograph (NIRSpec; [Jakobsen et al. \(2022\)](#)) on the James Webb Space Telescope (JWST; [Gardner et al. \(2023\)](#)) is a prime example where all of these challenges arise simultaneously: it critically undersamples the LSF across all gratings and observing modes, observations routinely span multiple gratings with complementary strengths (high-resolution to measure kinematics, low-resolution to constrain fluxes and continuum shapes), and systematic uncertainties in resolving power, absolute normalization, and wavelength solution have measurable impacts on the data ([de Graaff, Brammer, et al., 2025](#)).

unite is designed to address these challenges while providing a flexible, extensible, and reproducible framework for spectroscopic analysis applicable to any spectrograph, rigorously carrying instrumental systematics through to the final parameter uncertainties. By leveraging optimized libraries for probabilistic programming and automatic differentiation, unite delivers scientifically rigorous Bayesian inference without compromising on computational efficiency, enabling the analysis of large and mixed spectroscopic datasets with accurate error estimation.

Software design

Spectral fitting routines typically assume that the model evaluated at the pixel center is a good representation of the average of the model over the pixel domain, which is what the instrument actually records. This approximation is well justified when the spectrum is critically (Nyquist) sampled or over-sampled, i.e. when the signal changes slowly over the pixel domain, but breaks down when the spectrum is undersampled and the signal changes rapidly over the pixel domain. In the case of NIRSpec, a point source is undersampled by a factor of ~ 1.5 (de Graaff et al., 2024), leading to a bias of 10 – 20% in recovered line widths if not properly accounted for. This can be addressed by integrating the model over the pixel domain, providing the exact solution for the observed signal regardless of the degree of undersampling. We rely on the assumption that the LSF is well approximated by a Gaussian kernel, which is representative of the behaviour of NIRSpec (Shajib et al., 2025) and many other spectrographs, especially in the undersampled regime.

`unite` computes the integrals of continua and line models analytically where possible. However, analytic pixel integration is not possible for all model setups, in particular in the presence of optical-depth parametrized absorption lines where the nonlinear transmission $e^{-\tau\phi}$ couples the line depth and profile shape in a way that prevents closed-form solutions for the pixel integrals. In these cases, `unite` provides a numerical convolution mode which supersamples the intrinsic model on a fine wavelength grid and convolves with the wavelength-dependent LSF kernel to produce a pixel-convolved model. This model accurately captures the nonlinear coupling of line depth and profile shape, but is more computationally expensive than the analytic integration mode; users can choose the appropriate mode for their application.

`unite` is computationally efficient thanks to its JAX backend, which provides just-in-time (JIT) compilation and native GPU support. At its core, `unite` is a domain-specific language for building probabilistic models of spectroscopic data. Users build a declarative configuration of line and continuum components and assign priors to physical parameters via token instances that can be shared across multiple model components and combined arithmetically. The prior system is expressive: priors on any parameter can depend on other parameters through arithmetic expressions and topologically sorted dependency chains, enabling physical constraints such as requiring a broad component's velocity width to exceed that of a narrow component by a certain amount, or fixing flux ratios between doublet lines.

In addition, users specify the instrumental configuration carrying empirical calibrations of the wavelength-dependent resolving power, pixel scale, and flux normalization for each disperser, which can be shared across instruments. One aspect that sets `unite` apart from other spectral fitting tools is that it treats instrumental calibration parameters as first-class citizens in the inference process; priors can be specified on each of the aforementioned calibrations and are sampled jointly with astrophysical parameters, allowing for systematic instrumental uncertainties to directly propagate to the inferred properties. For example, when fitting NIRSpec data, users can incorporate the empirically measured wavelength and flux calibration offsets from de Graaff, Brammer, et al. (2025) as priors to obtain realistic uncertainty estimates on fluxes and kinematics by marginalizing over instrumental uncertainties.

All configurations are serializable to human-readable YAML for reproducibility and sharing. `unite` leverages NumPyro's probabilistic programming framework to assemble a model for inference with any compatible sampler, including SVI for quick exploratory fits, NUTS for full posterior sampling, and nested sampling for model comparison and evidence calculation. Finally, `unite` provides convenience functions for extracting results as parameter tables and per-spectrum model predictions into domain-appropriate FITS files, carrying physical units throughout, via `Astropy`.

The package is publicly available on GitHub and PyPI under the GPL-3.0-or-later license, with a DOI minted via Zenodo (Hvinding, 2026) and accompanied by Sphinx documentation including narrative guides, an API reference, and executable tutorials. CI/CD workflows ensure

93 that the code is tested and documented with each update, and the project is open to feedback
94 and contributions from the community.

95 State of the field

96 Spectral line analysis is among the most common operations in observational astronomy, and
97 the landscape of fitting software is correspondingly rich.

98 pPXF (Cappellari, 2017, 2023; Cappellari & Emsellem, 2004) is the standard tool for stellar
99 kinematics and stellar-population recovery, fitting observed galaxy spectra as non-negative
100 linear combinations of SSP templates and emission lines convolved with a parametric line-of-
101 sight velocity distribution (LOSVD), using an analytic Fourier representation of the LOSVD
102 to accurately recover kinematics even when the LOSVD is smaller than the instrumental
103 resolution.

104 PySpecKit (Ginsburg et al., 2022) is a general-purpose spectral fitting toolkit supporting a
105 range of profile shapes and optional MCMC uncertainty estimation.

106 LiMe (Fernández et al., 2024) is a modern library designed for large and complex datasets
107 including JWST spectra, with batch fitting across many spectra and flexible profile shapes.

108 Q3DFIT (Rupke, 2014; Rupke et al., 2021, 2023) targets IFU spectroscopy of quasar host
109 galaxies, fitting multi-component Gaussian profiles via least squares with the stellar continuum
110 pre-subtracted using pPXF.

111 BADASS (Sexton et al., 2021, 2024) is a comprehensive Bayesian emission-line code to
112 simultaneously infer AGN power-law continuum, FeII pseudo-continuum, stellar LOSVD (via
113 pPXF), and multi-component Gaussian emission lines in a single posterior.

114 Many of these codes rely on least-squares optimization (LMFIT (Newville et al., 2014) or
115 scipy.optimize (Virtanen et al., 2020)), which does not yield posterior distributions, or on
116 Bayesian inference via emcee (Foreman-Mackey et al., 2013) or pymc (Abril-Pla et al., 2023).
117 unite is more closely comparable to the latter category of Bayesian tools and builds natively
118 on JAX and NumPyro, inheriting JIT compilation, automatic differentiation, and GPU support,
119 enabling scalable inference across large spectroscopic samples.

120 Research impact statement

121 unite has already been used in several published JWST/NIRSpec analyses, demonstrating its
122 utility for emission line characterization.

- 123 ■ **Accurate Line Fluxes and Kinematics:** By accurately accounting for undersampling,
124 unite has been used to robustly characterize fluxes and kinematics for both emission
125 lines and absorption features in high-redshift galaxies, providing insights into the physical
126 processes driving star formation and black hole growth. (de Graaff, Hviding, et al., 2025;
127 Naidu et al., 2025; Sun et al., 2026; Wang et al., 2025, 2026)
- 128 ■ **Multi-Component Line Fitting:** unite has been used to identify broad Balmer emission
129 (broad $H\alpha$, $H\beta$) in high-redshift ($z \gtrsim 4$) galaxies observed with NIRSpec, providing
130 evidence for active supermassive black hole growth in the early universe. Individual
131 dispersers often lack sufficient signal-to-noise for such detections, making multi-disperser
132 joint fitting critical; by coupling flux constraints from low-resolution prism and kinematic
133 constraints from medium-resolution grating observations, unite enables robust detections
134 of these components. (de Graaff, Hviding, et al., 2025; Hviding et al., 2025, 2026)
- 135 ■ **Redshift Precision:** unite was used in the analysis of MoM-z14, which at the time of
136 publication was (and is currently) the most distant spectroscopically confirmed galaxy.
137 Even with faint, marginally detected emission lines, jointly fitting line parameters and

accounting for undersampling yielded a factor of 5 improvement in redshift precision over continuum estimates alone. (Naidu et al., 2026)

AI usage disclosure

Claude Code (Anthropic) was used as an AI coding assistant during the development of unite, its documentation, and the drafting of this paper. It was used to generate new code, refactor existing code, and write certain sections of the documentation. All scientific decisions, algorithmic design choices, and final content were made and verified by the authors.

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References

- Abril-Pla, O., Andreani, V., Carroll, C., Dong, L., Fomesbeck, C. J., Kochurov, M., Kumar, R., Lao, J., Luhmann, C. C., Martin, O. A., Osthege, M., Vieira, R., Wiecki, T., & Zinkov, R. (2023). PyMC: A modern and comprehensive probabilistic programming framework in Python. *PeerJ Computer Science*, 9(e1516). <https://doi.org/10.7717/peerj-cs.1516>
- Astropy Collaboration, Price-Whelan, A. M., Lim, P. L., Earl, N., Starkman, N., Bradley, L., Shupe, D. L., Patil, A. A., Corrales, L., Brasseur, C. E., Nöthe, M., Donath, A., Tollerud, E., Morris, B. M., Ginsburg, A., Vaher, E., Weaver, B. A., Tocknell, J., Jamieson, W., ... Astropy Project Contributors. (2022). The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package. 935(2), 167. <https://doi.org/10.3847/1538-4357/ac7c74>
- Astropy Collaboration, Price-Whelan, A. M., Sipőcz, B. M., Günther, H. M., Lim, P. L., Crawford, S. M., Conseil, S., Shupe, D. L., Craig, M. W., Dencheva, N., Ginsburg, A., VanderPlas, J. T., Bradley, L. D., Pérez-Suárez, D., de Val-Borro, M., Paper Contributors, (., Aldcroft, T. L., Cruz, K. L., Robitaille, T. P., ... Contributors, (. (2018). The Astropy Project: Building an Open-science Project and Status of the v2.0 Core Package. 156, 123. <https://doi.org/10.3847/1538-3881/aabc4f>
- Astropy Collaboration, Robitaille, T. P., Tollerud, E. J., Greenfield, P., Droettboom, M., Bray, E., Aldcroft, T., Davis, M., Ginsburg, A., Price-Whelan, A. M., Kerzendorf, W. E., Conley, A., Crighton, N., Barbary, K., Muna, D., Ferguson, H., Grollier, F., Parikh, M. M., Nair, P. H., ... Streicher, O. (2013). Astropy: A community Python package for astronomy. 558, A33. <https://doi.org/10.1051/0004-6361/201322068>
- Bingham, E., Chen, J. P., Jankowiak, M., Obermeyer, F., Pradhan, N., Karaletsos, T., Singh, R., Szerlip, P. A., Horsfall, P., & Goodman, N. D. (2019). Pyro: Deep universal probabilistic programming. *J. Mach. Learn. Res.*, 20, 28:1–28:6. <http://jmlr.org/papers/v20/18-403.html>
- Bradbury, J., Frostig, R., Hawkins, P., Johnson, M. J., Katariya, Y., Leary, C., Maclaurin, D., Necula, G., Paszke, A., VanderPlas, J., Wanderman-Milne, S., & Zhang, Q. (2018).

- 182 JAX: Composable transformations of Python+NumPy programs (Version 0.3.13). <http://github.com/jax-ml/jax>
- 183
- 184 Cappellari, M. (2017). Improving the full spectrum fitting method: accurate convolution with
- 185 Gauss-Hermite functions. *MNRAS*, 466(1), 798–811. [https://doi.org/10.1093/mnras/](https://doi.org/10.1093/mnras/stw3020)
- 186 [stw3020](https://doi.org/10.1093/mnras/stw3020)
- 187 Cappellari, M. (2023). Full spectrum fitting with photometry in PPXF: stellar population
- 188 versus dynamical masses, non-parametric star formation history and metallicity for 3200
- 189 LEGA-C galaxies at redshift $z \approx 0.8$. *MNRAS*, 526(3), 3273–3300. [https://doi.org/10.](https://doi.org/10.1093/mnras/stad2597)
- 190 [1093/mnras/stad2597](https://doi.org/10.1093/mnras/stad2597)
- 191 Cappellari, M., & Emsellem, E. (2004). Parametric Recovery of Line-of-Sight Velocity
- 192 Distributions from Absorption-Line Spectra of Galaxies via Penalized Likelihood. *PASP*,
- 193 116(816), 138–147. <https://doi.org/10.1086/381875>
- 194 de Graaff, A., Brammer, G., Weibel, A., Lewis, Z., Maseda, M. V., Oesch, P. A., Bezanson, R.,
- 195 Boogaard, L. A., Cleri, N. J., Cooper, O. R., Gottumukkala, R., Greene, J. E., Hirschmann,
- 196 M., Hviding, R. E., Katz, H., Labbé, I., Leja, J., Matthee, J., McConachie, I., ... Williams,
- 197 C. C. (2025). RUBIES: A complete census of the bright and red distant Universe with
- 198 JWST/NIRSpec. *A&A*, 697, A189. <https://doi.org/10.1051/0004-6361/202452186>
- 199 de Graaff, A., Hviding, R. E., Naidu, R. P., Greene, J. E., Miller, T. B., Leja, J., Matthee,
- 200 J., Brammer, G., Katz, H., Bezanson, R., Boogaard, L. A., Bose, S., Chisholm, J.,
- 201 Cleri, N. J., Dayal, P., Feldmann, R., Fudamoto, Y., Fujimoto, S., Furtak, L. J., ...
- 202 van der Wel, A. (2025). Little Red Dots host Black Hole Stars: A unified family of
- 203 gas-reddened AGN revealed by JWST/NIRSpec spectroscopy. *arXiv*, arXiv:2511.21820.
- 204 <https://doi.org/10.48550/arXiv.2511.21820>
- 205 de Graaff, A., Rix, H.-W., Carniani, S., Suess, K. A., Charlot, S., Curtis-Lake, E., Arribas,
- 206 S., Baker, W. M., Boyett, K., Bunker, A. J., Cameron, A. J., Chevallard, J., Curti,
- 207 M., Eisenstein, D. J., Franx, M., Hainline, K., Hausen, R., Ji, Z., Johnson, B. D., ...
- 208 Willott, C. (2024). Ionised gas kinematics and dynamical masses of $z \gtrsim 6$ galaxies from
- 209 JADES/NIRSpec high-resolution spectroscopy. *A&A*, 684, A87. [https://doi.org/10.1051/](https://doi.org/10.1051/0004-6361/202347755)
- 210 [0004-6361/202347755](https://doi.org/10.1051/0004-6361/202347755)
- 211 Fernández, V., Amorín, R., Firpo, V., & Morisset, C. (2024). LIME: A Line MEasuring library
- 212 for large and complex spectroscopic data sets. I. Implementation of a virtual observatory
- 213 for JWST spectra. *A&A*, 688, A69. <https://doi.org/10.1051/0004-6361/202449224>
- 214 Foreman-Mackey, D., Hogg, D. W., Lang, D., & Goodman, J. (2013). emcee: The MCMC
- 215 Hammer. *PASP*, 125(925), 306. <https://doi.org/10.1086/670067>
- 216 Gardner, J. P., Mather, J. C., Abbott, R., Abell, J. S., Abernathy, M., Abney, F. E., Abraham,
- 217 J. G., Abraham, R., Abul-Huda, Y. M., Acton, S., Adams, C. K., Adams, E., Adler, D.
- 218 S., Adriaensen, M., Aguilar, J. A., Ahmed, M., Ahmed, N. S., Ahmed, T., Albat, R., ...
- 219 Dellagatta, G. (2023). The James Webb Space Telescope Mission. *PASP*, 135(1048),
- 220 068001. <https://doi.org/10.1088/1538-3873/acd1b5>
- 221 Ginsburg, A., Sokolov, V., de Val-Borro, M., Rosolowsky, E., Pineda, J. E., Sipőcz, B. M., &
- 222 Henshaw, J. D. (2022). Pyspeckit: A Spectroscopic Analysis and Plotting Package. *AJ*,
- 223 163(6), 291. <https://doi.org/10.3847/1538-3881/ac695a>
- 224 Hviding, R. E. (2026). *Unite: Unified liNe integration turbo engine* (Version v2.9.2). Zenodo.
- 225 <https://doi.org/10.5281/zenodo.20158737>
- 226 Hviding, R. E., de Graaff, A., Liu, H., Goulding, A. D., Ma, Y., Greene, J. E., Boogaard, L.
- 227 A., Bunker, A. J., Cleri, N. J., Franx, M., Hirschmann, M., Leja, J., Matthee, J., Naidu,
- 228 R. P., Setton, D. J., Übler, H., Venturi, G., & Wang, B. (2026). The X-Ray Dot: Exotic
- 229 Dust or a Late-stage Little Red Dot? *ApJL*, 1000(1), L18. [https://doi.org/10.3847/2041-](https://doi.org/10.3847/2041-8213/ae4c88)
- 230 [8213/ae4c88](https://doi.org/10.3847/2041-8213/ae4c88)

- 231 Hviding, R. E., de Graaff, A., Miller, T. B., Setton, D. J., Greene, J. E., Labbé, I., Brammer,
232 G., Bezanson, R., Boogaard, L. A., Cleri, N. J., Leja, J., Maseda, M. V., McConachie, I.,
233 Matthee, J., Naidu, R. P., Oesch, P. A., Wang, B., Whitaker, K. E., & Williams, C. C.
234 (2025). RUBIES: A spectroscopic census of little red dots: All point sources with v-shaped
235 continua have broad lines. *A&A*, 702, A57. <https://doi.org/10.1051/0004-6361/202555816>
- 236 Jakobsen, P., Ferruit, P., Alves de Oliveira, C., Arribas, S., Bagnasco, G., Barho, R., Beck, T. L.,
237 Birkmann, S., Böker, T., Bunker, A. J., Charlot, S., de Jong, P., de Marchi, G., Ehrenwinkler,
238 R., Falcolini, M., Fels, R., Franx, M., Franz, D., Funke, M., ... Zincke, C. (2022). The
239 Near-Infrared Spectrograph (NIRSpec) on the James Webb Space Telescope. I. Overview
240 of the instrument and its capabilities. *A&A*, 661, A80. <https://doi.org/10.1051/0004-6361/202142663>
- 242 Naidu, R. P., Matthee, J., Katz, H., de Graaff, A., Oesch, P., Smith, A., Greene, J. E.,
243 Brammer, G., Weibel, A., Hviding, R., Chisholm, J., Labbé, I., Simcoe, R. A., Witten,
244 C., Atek, H., Baggen, J. F. W., Belli, S., Bezanson, R., Boogaard, L. A., ... Watson, D.
245 (2025). A “Black Hole Star” Reveals the Remarkable Gas-Enshrouded Hearts of the Little
246 Red Dots. *arXiv*, arXiv:2503.16596. <https://doi.org/10.48550/arXiv.2503.16596>
- 247 Naidu, R. P., Oesch, P. A., Brammer, G., Weibel, A., Li, Y., Matthee, J., Chisholm, J., Pollock,
248 C. L., Heintz, K. E., Johnson, B. D., Shen, X., Hviding, R. E., Leja, J., Tacchella, S.,
249 Ganguly, A., Witten, C., Atek, H., Belli, S., Bose, S., ... Whitaker, K. E. (2026). A Cosmic
250 Miracle: A Remarkably Luminous Galaxy at $z_{\text{spec}} = 14.44$ Confirmed with JWST. *OJAp*,
251 9, 56033. <https://doi.org/10.33232/001c.156033>
- 252 Newville, M., Stensitzki, T., Allen, D. B., & Ingargiola, A. (2014). *LMFIT: Non-linear*
253 *least-square minimization and curve-fitting for python* (Version 0.8.0). Zenodo. <https://doi.org/10.5281/zenodo.11813>
- 255 Phan, D., Pradhan, N., & Jankowiak, M. (2019). Composable effects for flexible and accelerated
256 probabilistic programming in NumPyro. *arXiv Preprint arXiv:1912.11554*.
- 257 Rupke, D. S. N. (2014). *IFSFIT: Spectral Fitting for Integral Field Spectrographs* (p.
258 ascl:1409.005).
- 259 Rupke, D. S. N., Schweitzer, M., Viola, V., Lutz, D., Sturm, E., Spoon, H., Veilleux, S., &
260 Kim, D.-C. (2021). *QUESTFIT: Fitter for mid-infrared galaxy spectra* (p. ascl:2112.002).
- 261 Rupke, D. S. N., Wylezalek, D., Zakamska, N., Veilleux, S., Vayner, A., Bertemes, C., Ishikawa,
262 Y., Liu, W., Lim, H. X. G., Murphree, G., Whitesell, L., McCrory, R., & Anicetti, C. (2023).
263 *q3dfit: PSF decomposition and spectral analysis for JWST-IFU spectroscopy*. Astrophysics
264 Source Code Library, record ascl:2310.004.
- 265 Sexton, R. O., Doan, S. M., Reefe, M., & Matzko, W. (2024). *BADASS: Bayesian AGN*
266 *Decomposition Analysis for SDSS Spectra*. Astrophysics Source Code Library, record
267 ascl:2412.004.
- 268 Sexton, R. O., Matzko, W., Darden, N., Canalizo, G., & Gorjian, V. (2021). Bayesian
269 AGN Decomposition Analysis for SDSS spectra: a correlation analysis of [O III] $\lambda 5007$
270 outflow kinematics with AGN and host galaxy properties. *MNRAS*, 500(3), 2871–2895.
271 <https://doi.org/10.1093/mnras/staa3278>
- 272 Shajib, A. J., Treu, T., Melo, A., Roberts-Borsani, G., Knabel, S., Cappellari, M., & Frieman, J.
273 A. (2025). An accurate measurement of the spectral resolution of the JWST Near Infrared
274 Spectrograph. *A&A*, 702, L12. <https://doi.org/10.1051/0004-6361/202556281>
- 275 Sun, W. Q., Naidu, R. P., Matthee, J., de Graaff, A., Chisholm, J., Greene, J. E., Oesch,
276 P. A., Torralba, A., Hviding, R. E., Brammer, G., Simcoe, R. A., Bose, S., Bouwens,
277 R., Dayal, P., Eilers, A.-C., Fei, Q., Furtak, L. J., Gottumukkala, R., Goulding, A., ...
278 Williams, C. C. (2026). Little Red Dot — Host Galaxy = Black Hole Star: A Gas-
279 Enshrouded Heart at the Center of Every Little Red Dot. *arXiv*, arXiv:2601.20929. <https://doi.org/10.48550/arXiv.2601.20929>

280 [//doi.org/10.48550/arXiv.2601.20929](https://doi.org/10.48550/arXiv.2601.20929)

281 Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D.,
282 Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson,
283 J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., ... SciPy
284 1.0 Contributors. (2020). SciPy 1.0: Fundamental Algorithms for Scientific Computing in
285 Python. *Nature Methods*, 17, 261–272. <https://doi.org/10.1038/s41592-019-0686-2>

286 Wang, B., Leja, J., Katz, H., Inayoshi, K., Cleri, N. J., de Graaff, A., Hviding, R. E., van
287 Dokkum, P., Greene, J. E., Labbé, I., Matthee, J., McConachie, I., Naidu, R. P., & Nelson,
288 E. J. (2025). The Missing Hard Photons of Little Red Dots: Their Incident Ionizing Spectra
289 Resemble Massive Stars. *arXiv*, arXiv:2508.18358. [https://doi.org/10.48550/arXiv.2508.](https://doi.org/10.48550/arXiv.2508.18358)
290 [18358](https://doi.org/10.48550/arXiv.2508.18358)

291 Wang, B., Leja, J., Labbe, I., Greene, J. E., Liu, H., de Graaff, A., Hviding, R. E., Matthee,
292 J., Quataert, E., Bezanson, R., Boogaard, L. A., Brammer, G., Burgasser, A. J., Chen,
293 Y.-X., Cleri, N. J., Cutler, S. E., Dayal, P., Furtak, L. J., Fujimoto, S., ... Zitrin, A. (2026).
294 Water absorption confirms cool atmospheres in two little red dots. *arXiv*, arXiv:2602.06024.
295 <https://doi.org/10.48550/arXiv.2602.06024>

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